Peer Review C3 - Problem 6.3

Given that 2 is a primitive root modulo 101, find $x \in \mathbb{Z}/101\mathbb{Z}$ such that $ord_{101}(x) = 10$.

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Solution

The modulus is $p = 101$, which is a prime number. The primitive root (admitted) is $g = 2$.

The order of the multiplicative group $\mathbb{Z}/101\mathbb{Z}^{\times}$ is $\phi(101) = 101 - 1 = 100$. Every element $x \in \mathbb{Z}/101\mathbb{Z}^{\times}$ can be written as a power of the primitive root g, i.e., $x = g^i = 2^i$ for some integer i.

The order of a power g^i is given by the formula:

$$ord_p(g^i) = \frac{p-1}{\gcd(p-1, i)}$$

In our case, we have $p = 101$ and we require $ord_{101}(x) = 10$. Substituting the known values:

$$10 = \frac{100}{\gcd(100, i)}$$

This implies that $\gcd(100, i)$ must be 10:

$$\gcd(100, i) = \frac{100}{10} = 10$$

We seek an integer i such that $\gcd(100, i) = 10$. Let $i = 10k$ for some integer k. Substituting this:

$$\gcd(100, 10k) = 10$$

Factoring out 10:

$$10 \cdot \gcd(10, k) = 10$$

Dividing by 10:

$$\gcd(10, k) = 1$$

We have to choose an integer k such that k is coprime to 10. The easiest number we can choose is $k = 1$, because $\gcd(10, 1) = 1$.

Thus, we set $k = 1$, which gives the exponent i:

$$i = 10k = 10 \cdot 1 = 10$$

The element x we are looking for is $x = 2^i$:

$$x = 2^{10} \pmod{101}$$

We calculate the value:

$$2^{10} = 1024$$

Now, we find $1024 \pmod{101}$:

$$1024 \equiv 14 \pmod{101}$$

$$x = 14$$

The required value is $\mathbf{x = 14}$.